Assignment 9
DATA 260

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Personal Configuration and Project Information

SID4	PORT_BASE	PREFIX	SEED	VERIFY_SEED	DOMAIN_ID
4098	8498	s4098	4098	264098	2

Item	Value
Domain	Municipal transit incidents (main entity: Incident)
Hardware	Apple MacBook Pro, M5 chip, 24 GB RAM
Local model	qwen3:8b with Ollama, called through my src/model_client.py
Cache	Redis Stack 7.4.0-v3 in Docker (s4098-redis, port 6380)
Embeddings	sentence-transformers/all-MiniLM-L6-v2
Database	s4098_rel (my MySQL database from HW4/HW5)



Figure A. Redis Stack, MySQL and qwen3:8b running.

I kept working in the same repository as HW1-HW8. The new code is in `src/hw9_cache/`, `src/hw9_agents/` and `code/hw9/`. The results are in `reports/hw09/`. All Redis keys start with `s4098:`.

Part 1 - Redis View Counter

I added the method `increment_incident_views(incident_id)`. Every time an incident is viewed, it adds 1 to the Redis key `s4098:views:incident:<id>` with INCR. Every 10 views, it saves the count to a new column `view_count` in the incidents table of s4098_rel and prints "Synced views to database". Then it returns the count.

```
def increment_incident_views(self, incident_id: int) -> int:
    key = self.views_key(incident_id)          # s4098:views:incident:<id>
    count = self.redis_client.incr(key)        # add 1 in Redis
    if count % self.sync_every == 0:           # every 10 views
        with self.engine.begin() as conn:
            conn.execute(text("UPDATE incidents SET view_count = :c WHERE id = :id"),
                         {"c": count, "id": incident_id})
        print("✓ Synced views to database")
    return count
```

```
Kazi Samina Maraj Mumu | part1_view_counter.py (Part 1) | PREFIX=s4098 REDIS=redis://localhost:6380/0
Added column incidents.view_count (INT NOT NULL DEFAULT 0)
Incident INC-000042 (id=42): Transit Infrastructure Issue Incident 41
Redis key: s4098:views:incident:42

=== Testing Counter ===

Access #1: Total = 1
Access #2: Total = 2
Access #3: Total = 3
Access #4: Total = 4
Access #5: Total = 5
Access #6: Total = 6
Access #7: Total = 7
Access #8: Total = 8
Access #9: Total = 9
Access #10: Total = 10
✓ Synced views to database   (UPDATE incidents SET view_count=10 WHERE id=42)
Access #11: Total = 11
Access #12: Total = 12
Access #13: Total = 13
Access #14: Total = 14
Access #15: Total = 15
Access #16: Total = 16
Access #17: Total = 17
Access #18: Total = 18
Access #19: Total = 19
Access #20: Total = 20
✓ Synced views to database   (UPDATE incidents SET view_count=20 WHERE id=42)
Access #21: Total = 21
Access #22: Total = 22
Access #23: Total = 23
Access #24: Total = 24
Access #25: Total = 25

=== After the run ===
Redis s4098:views:incident:42 = 25
s4098_rel incidents.view_count (id=42) = 20
DB writes: 2 for 25 views

=== Timing: 200 views, Redis path vs DB on every view ===
Redis INCR (+ DB write every 10th): mean 0.242 ms, p50 0.128 ms
DB UPDATE + SELECT every view:      mean 1.141 ms, p50 1.112 ms

saved /Users/samina/Documents/data260-4098/reports/hw09/raw/part1_counter.json
((.venv) ) samina@Kazis-MacBook-Pro-2 data260-4098 %
```

Figure 1. 25 views of INC-000042. The count was saved to the database at 10 and 20.

After 25 views, Redis had 25 and the database had 20, which is correct because the next save happens at 30. I also timed 200 views each way: the Redis counter took 0.242 ms per view on average, and updating the database on every view took 1.141 ms. So Redis was about 4.7 times faster and the database only got 2 writes instead of 25.

Part 2 - Semantic Response Cache

I. How it works

The cache sits in front of my HW8 agents (Planner, Writer, Reviewer). When a question comes in:

- I turn the question into an embedding and search Redis for the most similar saved question (cosine similarity, vector index `s4098:idx:semcache`).
- If the similarity is 0.85 or more, it is a **cache hit**: I return the saved answer with is_cached=True and the similarity score.
- Otherwise it is a **cache miss**: I get the facts for the question from s4098_rel, run the HW8 agents with qwen3:8b, and save the question, embedding, answer and a data-version marker in Redis. I return is_cached=False.

I record the time before and after each step to measure latency. On a miss, the agents ran in-process (the same HW8 Planner, Writer and Reviewer functions called directly, not through my HW8 Kafka pipeline). The test script called `qa.ask()` directly; the same method is also exposed as the `/qa` endpoint in `src/hw9_cache/api.py`.

```
VectorField("embedding", "FLAT",
            {"TYPE": "FLOAT32", "DIM": 384, "DISTANCE_METRIC": "COSINE"})
q = Query("*=>[KNN 1 @embedding $vec AS dist]").sort_by("dist").dialect(2)
similarity = 1.0 - float(doc.dist)     # Redis returns the cosine distance
```

```
hit, nearest = self.cache.lookup(vector)        # KNN-1 + threshold check
if hit is not None:                              # cache hit
    return {"answer": hit.answer, "is_cached": True, "similarity": hit.similarity}
grounding = self.facts.ground(question)          # cache miss: facts from s4098_rel
agent_out = self.agent.run(question, grounding.as_prompt())   # HW8 agents
self.cache.store(question, vector, agent_out["answer"],
                 grounding.data_version, grounding.deps)
return {"answer": agent_out["answer"], "is_cached": False, "similarity": best_sim}
```

II. Results

I tested 12 questions: 5 new ones, 3 exact duplicates and 4 paraphrases.

```
Kazi Samina Maraj Mumu | part2_semantic_cache.py (Part 2) | PREFIX=s4098 REDIS=redis://localhost:6380/0
threshold=0.85  embedder=sentence-transformers/all-MiniLM-L6-v2  agent path=inproc  (flushed 0 old cache entries)

--- #01 [new] How many delay incidents were recorded on VTA Route 22?
2026-11-06T03:51:14.211Z [QA] CACHE MISS best_similarity=0.0000 < 0.85
2026-11-06T03:51:27.373Z [QA] agent answered in 13124.91 ms; stored s4098:semcache:33a57fe2b033 data_version=276681c63abb; latency=13271.23 ms
    is_cached=False  similarity=0.0000  latency=13271.2 ms
    answer: The route code for VTA Route 22 is RT-003, which has 266 recorded delay incidents according to the s4098_rel transit incident database. This matche...

--- #02 [new] Who operates VTA Route 522?
2026-11-06T03:51:27.590Z [QA] CACHE MISS best_similarity=0.5362 < 0.85
2026-11-06T03:51:35.786Z [QA] agent answered in 8182.62 ms; stored s4098:semcache:6cfbce391810 data_version=40ea672fa3da; latency=8411.45 ms
    is_cached=False  similarity=0.5362  latency=8411.5 ms
    answer: The s4098_rel transit incident database indicates that Route RT-004 is 'VTA Route 522' and is operated by VTA. Therefore, VTA Route 522 is operated...

--- #03 [exact duplicate of #1] How many delay incidents were recorded on VTA Route 22?
2026-11-06T03:51:35.907Z [QA] CACHE HIT  similarity=1.0000 >= 0.85 matched="How many delay incidents were recorded on VTA Route 22?" latency=120.9 ms
    is_cached=True  similarity=1.0000  latency=120.9 ms  matched="How many delay incidents were recorded on VTA Route 22?"
    answer: The route code for VTA Route 22 is RT-003, which has 266 recorded delay incidents according to the s4098_rel transit incident database. This matche...

--- #04 [paraphrase of #1] What is the number of delay incidents logged for VTA Route 22?
2026-11-06T03:51:35.982Z [QA] CACHE HIT  similarity=0.9375 >= 0.85 matched="How many delay incidents were recorded on VTA Route 22?" latency=75.22 ms
    is_cached=True  similarity=0.9375  latency=75.2 ms  matched="How many delay incidents were recorded on VTA Route 22?"
    answer: The route code for VTA Route 22 is RT-003, which has 266 recorded delay incidents according to the s4098_rel transit incident database. This matche...

--- #05 [new] What category is incident INC-000042?
2026-11-06T03:51:35.990Z [QA] CACHE MISS best_similarity=0.3087 < 0.85
2026-11-06T03:51:44.967Z [QA] agent answered in 8970.69 ms; stored s4098:semcache:82d8f17f868f data_version=84d5c63191bb; latency=8984.46 ms
    is_cached=False  similarity=0.3087  latency=8984.5 ms
    answer: The incident code INC-000042 is categorized as 'infrastructure-issue' according to the s4098_rel transit incident database. This incident, labeled ...

--- #06 [paraphrase of #2] Which operator runs VTA Route 522?
2026-11-06T03:51:45.132Z [QA] CACHE HIT  similarity=0.8854 >= 0.85 matched="Who operates VTA Route 522?" latency=165.02 ms
    is_cached=True  similarity=0.8854  latency=165.0 ms  matched="Who operates VTA Route 522?"
    answer: The s4098_rel transit incident database indicates that Route RT-004 is 'VTA Route 522' and is operated by VTA. Therefore, VTA Route 522 is operated...

--- #07 [exact duplicate of #5] What category is incident INC-000042?
2026-11-06T03:51:45.144Z [QA] CACHE HIT  similarity=1.0000 >= 0.85 matched="What category is incident INC-000042?" latency=12.34 ms
    is_cached=True  similarity=1.0000  latency=12.3 ms  matched="What category is incident INC-000042?"
    answer: The incident code INC-000042 is categorized as 'infrastructure-issue' according to the s4098_rel transit incident database. This incident, labeled ...

--- #08 [new] How many incidents in total were recorded on VTA Route 55?
2026-11-06T03:51:45.281Z [QA] CACHE MISS best_similarity=0.8203 < 0.85
2026-11-06T03:51:55.525Z [QA] agent answered in 10234.62 ms; stored s4098:semcache:b3d1359a9c20 data_version=d553a0c95c65; latency=10380.25 ms
    is_cached=False  similarity=0.8203  latency=10380.2 ms
    answer: The total number of incidents recorded on VTA Route 55 (Route RT-005) is 993. This includes 264 collisions, 223 delays, 252 infrastructure issues, ...

--- #09 [paraphrase of #8] What is the total incident count for VTA Route 55?
2026-11-06T03:51:55.654Z [QA] CACHE HIT  similarity=0.9118 >= 0.85 matched="How many incidents in total were recorded on VTA Route 55?" latency=129.12 ms
    is_cached=True  similarity=0.9118  latency=129.1 ms  matched="How many incidents in total were recorded on VTA Route 55?"
    answer: The total number of incidents recorded on VTA Route 55 (Route RT-005) is 993. This includes 264 collisions, 223 delays, 252 infrastructure issues, ...

--- #10 [new (unrelated)] What is the capital of Australia?
2026-11-06T03:51:55.785Z [QA] CACHE MISS best_similarity=0.1320 < 0.85
2026-11-06T03:52:01.790Z [QA] agent answered in 6000.82 ms; stored s4098:semcache:da36863872e2 data_version=44136fa355b3; latency=6136.12 ms
    is_cached=False  similarity=0.1320  latency=6136.1 ms
    answer: The question about the capital of Australia is unrelated to municipal transit incidents. The s4098_rel transit incident database has no record for ...
```

Figure 2. Cache misses and hits with similarity and latency.

```
--- #11 [exact duplicate of #2] Who operates VTA Route 522?
2026-11-06T03:52:01.952Z [QA] CACHE HIT  similarity=1.0000 >= 0.85 matched="Who operates VTA Route 522?" latency=161.76 ms
    is_cached=True  similarity=1.0000  latency=161.8 ms  matched="Who operates VTA Route 522?"
    answer: The s4098_rel transit incident database indicates that Route RT-004 is 'VTA Route 522' and is operated by VTA. Therefore, VTA Route 522 is operated...

--- #12 [paraphrase of #5] Which incident category does INC-000042 belong to?
2026-11-06T03:52:01.963Z [QA] CACHE HIT  similarity=0.9522 >= 0.85 matched="What category is incident INC-000042?" latency=11.57 ms
    is_cached=True  similarity=0.9522  latency=11.6 ms  matched="What category is incident INC-000042?"
    answer: The incident code INC-000042 is categorized as 'infrastructure-issue' according to the s4098_rel transit incident database. This incident, labeled ...

=== Summary table ===
  # kind                    is_cached similarity  latency ms
  1 new                     False         0.0000     13271.2
  2 new                     False         0.5362      8411.5
  3 exact duplicate of #1   True          1.0000       120.9
  4 paraphrase of #1        True          0.9375        75.2
  5 new                     False         0.3087      8984.5
  6 paraphrase of #2        True          0.8854       165.0
  7 exact duplicate of #5   True          1.0000        12.3
  8 new                     False         0.8203     10380.2
  9 paraphrase of #8        True          0.9118       129.1
 10 new (unrelated)         False         0.1320      6136.1
 11 exact duplicate of #2   True          1.0000       161.8
 12 paraphrase of #5        True          0.9522        11.6

Cache hit rate: 7/12 = 58.3%
Average latency cached:     96.56 ms (7 queries)
Average latency non-cached: 9436.7 ms (5 queries)
Observed ratio non-cached / cached: 97.7x

saved /Users/samina/Documents/data260-4098/reports/hw09/raw/part2_results.json
((.venv) ) samina@Kazis-MacBook-Pro-2 data260-4098 % echo "Kazi Samina Maraj Mumu"
Kazi Samina Maraj Mumu
((.venv) ) samina@Kazis-MacBook-Pro-2 data260-4098 %
```

Figure 3. Summary of the 12 questions.

Result	Value
Cache hit rate	7/12 = 58.3%
Average latency, cached	96.56 ms
Average latency, not cached	9436.7 ms

All duplicates and paraphrases were hits and all new questions were misses. The lowest paraphrase similarity was 0.8854, which is very close to 0.85, so I tested other thresholds in Part 3.

Part 3 - Threshold Economics and Stale Answers

3.1 Labeled test set

I made `reports/hw09/cases/cache_cases.yaml` with 40 labeled questions: 10 exact duplicates (should hit), 10 paraphrases (should hit), 10 questions that look similar but have a different answer (should miss), and 10 unrelated questions (should miss). I committed it before running the sweep and did not change it.

One sample case from each group (from cache_cases.yaml):

Group	Expected	Seed (cached question)	Test question
exact	hit	Who operates VTA Route 522?	Who operates VTA Route 522?
paraphrase	hit	Who operates VTA Route 522?	Which operator runs VTA Route 522?
similar_different	miss	How many delay incidents were recorded on VTA Route 22?	How many collision incidents were recorded on VTA Route 22?
unrelated	miss	(none)	What is the capital of Australia?

Labeling mistake: I wrote the labels using my HW4 seed script, but the route codes and incident codes in my real database are different. When I checked every pair against the database, C24 (INC-000043 vs INC-000042) turned out to have the same category, so it is not really a "different answer" for that question. I did not change the file. I wrote this in `reports/hw09/LABELING_NOTES.md` and still counted C24 as a false hit, because the cached answer names the wrong incident (INC-000042). Removing C24 would not change my threshold choice, since it is a false hit at every threshold.

3.2 Threshold sweep

For each threshold I emptied the cache, saved the same 10 starting questions, and ran all 40 cases. False-hit rate = false hits / hits.

Threshold	Hit rate	False-hit rate	Mean latency (ms)	Total cost
0.75	75.0%	33.3%	2067.8	210
0.80	70.0%	28.6%	2530.7	172
0.85	70.0%	28.6%	2435.4	172
0.90	60.0%	29.2%	3392.3	156
0.95	40.0%	18.8%	4887.9	84

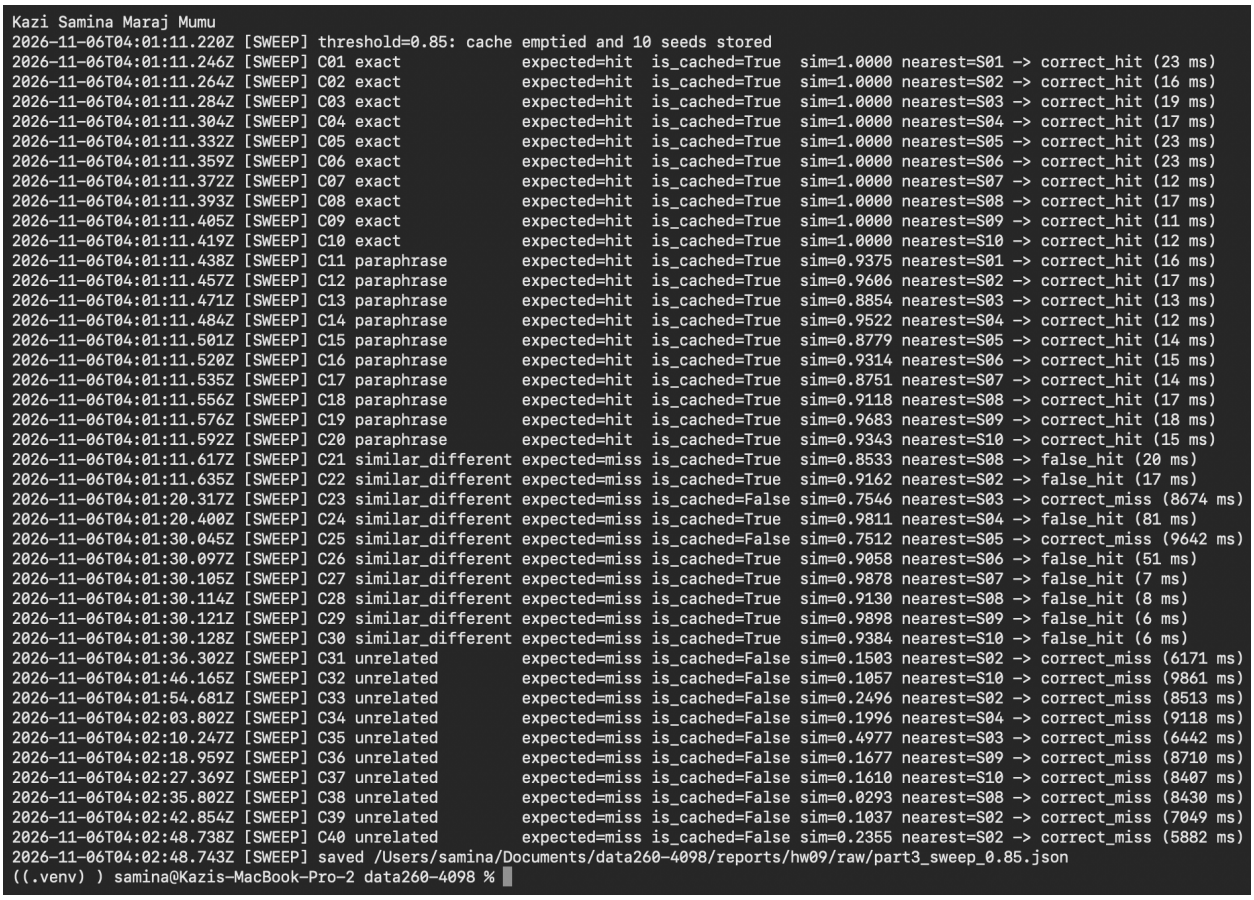


Figure 4. All 40 cases at threshold 0.85.

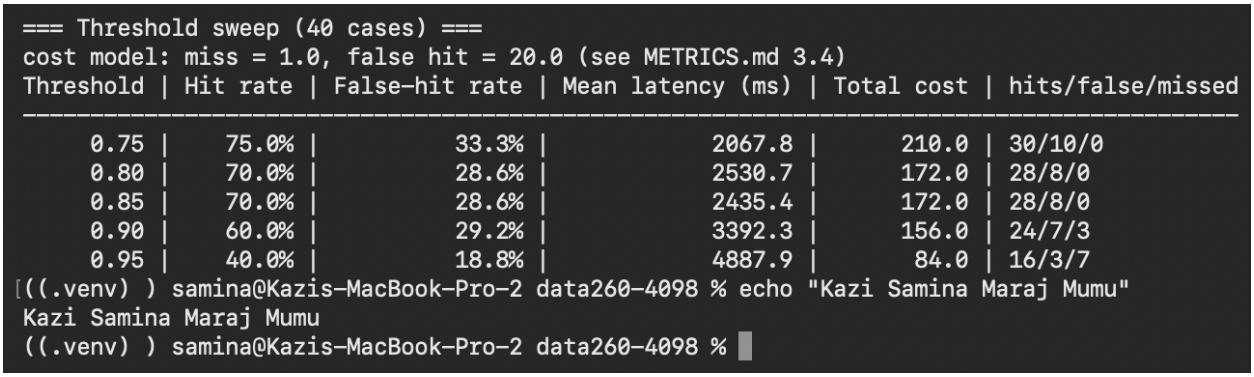


Figure 5. Sweep results for all thresholds.

Unrelated questions always missed. The problem was the similar-looking questions: their similarity (0.75-0.99) overlapped with the paraphrases (0.88-0.97). At 0.85, 8 of the 10 similar-looking questions got a wrong cached answer. At 0.95 only 3 did, but 7 of the 10 paraphrases also stopped hitting, and the average latency went up because more questions had to run the agents.

3.3 One false hit in full

Item	Value
Cached query	What is the title of incident INC-000250?
Incoming query	What is the title of incident INC-000251?
Similarity	0.9898
Cached answer returned	The title of incident INC-000250 is "Transit Service Suspension Incident 249".
Correct answer	INC-000251 is titled "Transit Infrastructure Issue Incident 250".

Why this happened: the two questions are exactly the same except for one digit in the incident code. The embedding model looks at the meaning of the whole sentence, and almost all of the words are the same, so the vectors are almost the same. The model does not treat the incident number as important. The other false hits at 0.95 were the same kind of question (INC-000042/043 and INC-000122/123).

3.4 Cost model and threshold choice

- **Cache miss = 1.** A miss runs the three agents, which took about 9.4 s on average in Part 2. The user waits, but the answer is freshly generated from the current data.
- **False hit = 20.** A false hit gives a wrong answer that looks correct. Someone might act on it, and fixing it needs a person to notice, check the database and ask again. I estimated that as about 3 minutes, which is around 20 misses worth of waiting.

Threshold	Misses	False hits	Total cost
0.75	10	10	10 + 200 = 210
0.80	12	8	12 + 160 = 172
0.85	12	8	12 + 160 = 172
0.90	16	7	16 + 140 = 156
0.95	24	3	24 + 60 = 84

I would ship 0.95 because it has the lowest total cost. 0.95 has the lowest cost whenever a false hit costs more than 2.4 misses (compared with 0.85: 12 more misses but 5 fewer false hits). The downside is a lower hit rate (40%) and higher latency. Since 0.95 still had 3 false hits on questions with different incident numbers, a better fix would be to also check that the incident code or route in the new question matches the cached one.

3.5 Stale answers and invalidation

When I save an answer, I also save which database rows it used and a data_version, which is a hash of those rows. With invalidation turned on, every cache hit reads those rows again. If the hash is different, the data changed, so I delete the old entry and run the agents again.

```
current = self.facts.load_facts(hit.deps).data_version   # read the rows again
if current != hit.data_version:                           # data changed
    self.cache.backend.delete(hit.key)                    # remove stale answer
    hit = None                                            # run the agents again
```

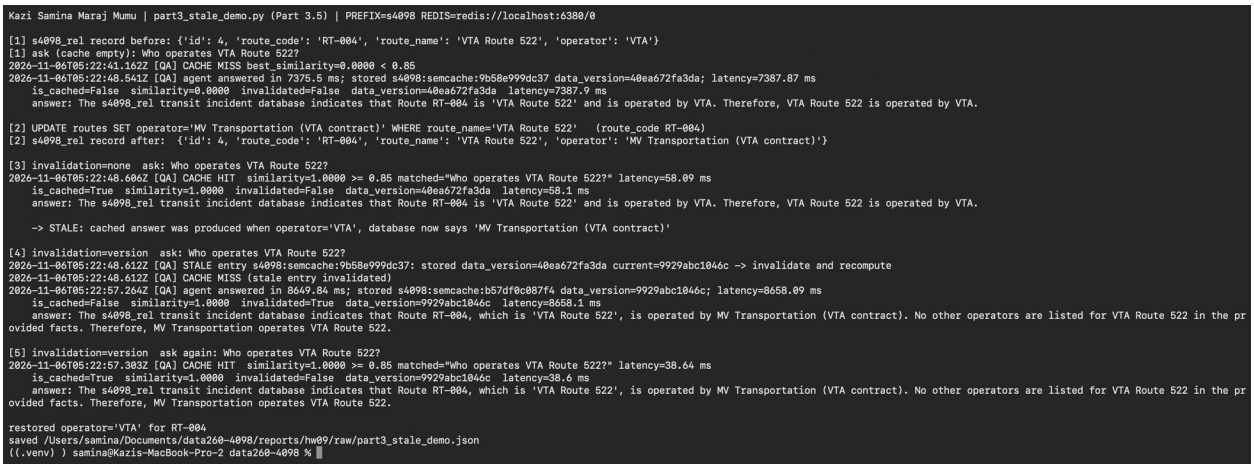


Figure 6. Stale answer and invalidation demo.

I asked "Who operates VTA Route 522?" (answer: VTA). Then I changed the operator of that route in s4098_rel to "MV Transportation (VTA contract)". Without invalidation, the cache still answered VTA, which is the stale hit. With invalidation, the data_version changed from 40ea672fa3da to 9929abc1046c, so the old entry was deleted and the new answer said MV Transportation. The next question was a hit with the new answer.

Trade-off: this always catches changes to the rows the answer used, but every cache hit needs one extra database query, so hits are a little slower. It also only works if I saved the right rows for each answer.

Part 4 - Self-Improving Multi-Agent System

I. Datasets

All files are in `code/hw9/part4_datasets/`.

File	What it has
reference_facts.json	19 correct facts about my transit domain (the ground truth for the Evaluator)
training_data.json	10 questions used to create lessons
validation_data.json	10 questions used to decide if a lesson is kept
test_data.json	12 questions used only for the final test
lessons.json	the saved lessons

The question files only have the questions, with no answers. Each split has 2 questions outside my domain, like "Who won the 2022 FIFA World Cup?", to check that the system does not answer things it has no facts for.

II. The two agents

- **Answering Agent:** answers the question with qwen3:8b. If a lesson or the feedback tells it to, it looks up the reference facts first.
- **Evaluator Agent:** checks the answer against the reference facts. It checks that the important words or numbers are in the answer, and qwen3:8b also judges the answer. It returns PASS or REVISE, an explanation and feedback. The feedback hides the exact number so the Answering Agent has to look it up.

Answering Agent (`AnsweringAgent.answer()`, short version):

```
refs = lookup(question) if use_lookup else None      # reference lookup tool
system = f"You are the Answering Agent for {DOMAIN}. Answer ... in at most 60 words."
if lessons:                                          # loaded lessons go in the prompt
    system += "Lessons from earlier evaluated mistakes ..." + lesson texts
user = f"Question: {question}"
if refs is not None:
    user += "Reference dataset lookup (agency facts): ..." + fact texts
if previous is not None:                             # revision attempt
    user += f"Your previous answer: {previous} Evaluator feedback: {feedback}"
text = _llm(system, user, json_mode=False)           # qwen3:8b
```

Evaluator Agent (`EvaluatorAgent.evaluate()`, short version):

```
fact = fact_for_topic(topic) if topic != "out_of_scope" else None
if fact is None:
    rule_pass = declines(answer)                  # out of scope: must decline
else:
    missing = missing_terms(answer, fact)         # required words/numbers present?
    rule_pass = not missing
raw = _llm("You are the Evaluator Agent. Judge the answer ONLY against the ground truth"
           ... 'Return ONLY JSON: {"verdict", "explanation", "feedback"}', ...)
result = "PASS" if rule_pass and llm_verdict == "PASS" else "REVISE"
feedback = self._mask(feedback, fact)             # hide the exact value
```

The revision loop:

```
for n in range(1, MAX_ATTEMPTS + 1):          # MAX_ATTEMPTS = 2
    ans = self.answering.answer(q, lessons=lessons, use_lookup=bool(lessons) or n > 1,
                                previous=previous, feedback=feedback)
    ev = self.evaluator.evaluate(q, topic, ans["answer"])
    if ev["result"] == "PASS" or not self.repair:
        break
    previous, feedback = ans["answer"], ev["feedback"]
```

III. 4.1 Repair using evaluator feedback

For the controlled test, I gave the system an incomplete first answer on purpose.

```
Kazi Samina Maraj Mumu | part4_self_improving.py demo-revision | model=qwen3:8b via model_client

=== 4.1 Controlled revision case ===
Question: When should a bus bridge be requested during a light-rail service suspension?
Controlled initial answer (intentionally incomplete, no time limit): A bus bridge can be requested when a light-rail suspension is long.

  attempt 1: A bus bridge can be requested when a light-rail suspension is long.
    evaluator: REVISE (missing_value) The answer is too vague and does not specify the required duration for requesting a bus bridge.
    feedback: Specify that a bus bridge should be requested when the light-rail suspension is expected to last more than [value] minutes. Check the reference dataset before a
  attempt 2: A bus bridge should be requested when a light-rail suspension is expected to last more than 20 minutes.
    evaluator: PASS (none) The answer correctly states the condition under which a bus bridge should be requested.

--- Revision sequence ---
Attempt 1 answer: A bus bridge can be requested when a light-rail suspension is long.
  Evaluator result: REVISE  failure_type=missing_value  rule_pass=False  llm_verdict=REVISE
  Explanation: The answer is too vague and does not specify the required duration for requesting a bus bridge.
  Feedback: Specify that a bus bridge should be requested when the light-rail suspension is expected to last more than [value] minutes. Check the reference dataset before answering.
Attempt 2 answer: A bus bridge should be requested when a light-rail suspension is expected to last more than 20 minutes.
  reference lookup used: ['bus_bridge', 'city_notification']
  Evaluator result: PASS  failure_type=none  rule_pass=True  llm_verdict=PASS
  Explanation: The answer correctly states the condition under which a bus bridge should be requested.

Number of attempts: 2 (max 2)
Final evaluation:   PASS
Final response:     A bus bridge should be requested when a light-rail suspension is expected to last more than 20 minutes.
saved /Users/samina/Documents/data260-4098/reports/hw09/raw/part4_revision_demo.json
((.venv) ) samina@Kazis-MacBook-Pro-2 data260-4098 %
```

Figure 7. Full revision sequence.

Step	Content
Question	When should a bus bridge be requested during a light-rail service suspension?
Initial answer	A bus bridge can be requested when a light-rail suspension is long.
Evaluator feedback	REVISE: the answer does not give the time limit. "Specify ... more than [value] minutes. Check the reference dataset before answering."
Revised answer	A bus bridge should be requested when a light-rail suspension is expected to last more than 20 minutes.
Final evaluation	PASS
Number of attempts	2

IV. 4.2 Saving and reusing lessons

When the same type of mistake happened twice in training, the Evaluator wrote a lesson and saved it in lessons.json. Before answering, the Answering Agent loads any lesson whose keywords match the question.

```
Kazi Samina Maraj Mumu
  ** LESSON CREATED lesson_001 for repeated failure 'missing_value' (seen in ['tr01', 'tr03'])
     failure_pattern: Answers provide incorrect or incomplete information without aligning with the specific requirements or thresholds outlined in the ground truth.
     correction:      Ensure answers strictly adhere to the exact criteria or timeframes provided in the ground truth, avoiding assumptions or generalizations.
     trigger:         {'type': 'keywords', 'keywords': ['bridge', 'infrastructure-issue', 'maintenance', 'opened', 'requested', 'service', 'soon', 'suspension', 'ticket']}
--
  ** LESSON CREATED lesson_002 for repeated failure 'wrong_label' (seen in ['tr04', 'tr09'])
     failure_pattern: The answer does not align with the specific labels or categories defined in the ground truth.
     correction:      Ensure that the answer strictly follows the exact labels and categories provided in the ground truth without adding, removing, or modifying any elements.
     trigger:         {'type': 'keywords', 'keywords': ['allowed', 'category', 'field', 'required', 'submitting']}
((.venv) ) samina@Kazis-MacBook-Pro-2 data260-4098 %
```

Figure 8. Two lessons created during training.

```
Kazi Samina Maraj Mumu
=== 4.2 Lesson reuse on a later related question ===
saved lesson_001 (kept, created 2026-11-05 from ['tr01', 'tr03'], use_count=4): Ensure answers strictly adhere to the exact criteria or timeframes provided in the ground truth, avoiding assumptions or generalizati
ons.
saved lesson_002 (kept, created 2026-11-05 from ['tr04', 'tr09'], use_count=0): Ensure that the answer strictly follows the exact labels and categories provided in the ground truth without adding, removing, or mod
ifying any elements.

New question va01: Within how many minutes must a rider alert be posted after a service suspension is confirmed?
  LESSON LOADED lesson_001 (missing_value; matched keywords: service, suspension)
  attempt 1: The rider alert must be posted within 10 minutes after confirming a service suspension.
    evaluator: PASS (none) The answer correctly states the required time frame.

Lessons loaded before answering: ['lesson_001']
Reference lookup on attempt 1: ['rider_alert', 'bus_bridge']
Result: PASS on attempt 1, final PASS after 1 attempt(s)
Answer: The rider alert must be posted within 10 minutes after confirming a service suspension.
use_count now: lesson_001=5, lesson_002=0
saved /Users/samina/Documents/data260-4098/reports/hw09/raw/part4_reuse_demo.json
((.venv) ) samina@Kazis-MacBook-Pro-2 data260-4098 %
```

Figure 9. lesson_001 loaded for a new question (va01) and used.

lesson_001 was created from tr01 and tr03, which both missed the required time limit (tr01 gave no time limit and tr03 said 15 minutes instead of 4 hours). Later, the new question va01 ("Within how many minutes must a rider alert be posted...") loaded lesson_001, looked up the reference and passed on the first attempt. Its use_count went from 4 to 5.

V. 4.3 Validation

I tested each lesson on the validation set before and after adding it. I kept it only if the failures did not get worse, no previously passing answer failed the evaluation after adding it, and no out-of-domain question got an answer it should not get.

```
Kazi Samina Maraj Mumu
  GATE lesson_001: first-attempt failures 10 -> 8, regressions=[], unsupported out-of-scope 0 -> 0  => KEEP
  GATE lesson_002: first-attempt failures 8 -> 8, regressions=[], unsupported out-of-scope 0 -> 0  => KEEP
Kept lessons: ['lesson_001', 'lesson_002']
((.venv) ) samina@Kazis-MacBook-Pro-2 data260-4098 %
```

Figure 10. Validation run: both lessons passed the three checks on validation_data.json.

Lesson	First-try failures (before -> after)	Regressions	Decision
lesson_001	10 -> 8	none	Keep
lesson_002	8 -> 8	none	Keep (not really tested)

lesson_002 passed only because no validation question loaded it, so the validation set did not really test it. Final validation accuracy stayed at 9/10 before and after, because va09 (passengersAffected when unknown) failed in both runs.

Extra check for lesson_002. To test it properly, I wrote 5 new questions that are not in the training, validation or test files and that do load lesson_002: three about field names, category labels and note types, and two out-of-domain questions that contain the word "allowed". I ran each one without and with lesson_002 and applied the same three checks (`code/hw9/part4_extra_lesson_check.py`).

```
Kazi Samina Maraj Mumu
=== Extra check of lesson_002 on 5 new questions ===
id   topic              lesson_002 loaded  before (1st/final, tries)   after (1st/final, tries)
x1   required_fields    True               REVISE/PASS, 2              PASS/PASS, 1
x2   valid_categories   True               REVISE/PASS, 2              PASS/PASS, 1
x3   related_types      True               REVISE/PASS, 2              PASS/PASS, 1
x4   out_of_scope       True               REVISE/PASS, 2              REVISE/REVISE, 2
x5   out_of_scope       True               REVISE/REVISE, 2            REVISE/PASS, 2

First-try correct: 0/5 -> 3/5
Final correct:     4/5 -> 4/5
Average attempts:  2.00 -> 1.40
Checks: {'failures_improve_or_same': True, 'no_regression_on_correct': False, 'no_new_unsupported_answers': True}
Result: lesson_002 FAILS (would be rejected)
((.venv) ) samina@Kazis-MacBook-Pro-2 data260-4098 %
```

Figure 11. Extra check: 5 new questions that load lesson_002, before and after adding it.

Question	Without lesson_002 (1st / final)	With lesson_002 (1st / final)
x1 required fields	REVISE / PASS	PASS / PASS
x2 category labels	REVISE / PASS	PASS / PASS
x3 note types	REVISE / PASS	PASS / PASS
x4 "Which pets are allowed on trains?"	REVISE / PASS	REVISE / REVISE
x5 "Is food allowed on light-rail trains?"	REVISE / REVISE	REVISE / PASS

lesson_002 helped the questions it was made for: all three label questions passed on the first try (first-try correct 0/5 -> 3/5). But x4 passed the automated evaluation without the lesson and failed it with the lesson, so the check "no previously passing answer fails" did not hold and **I rejected lesson_002** (status "rejected" in lessons.json). When I read the answers, the x4 answer with the lesson ("The question cannot be answered based on available reference data.") was actually a refusal that my Evaluator's refusal rule did not recognize. But the lesson also hurt x5: its own instruction text ended up in the answer ("The answer must strictly follow the exact labels and categories..."). Because of that, I kept the rejection.

VI. 4.4 Four variants on the test set

After rejecting lesson_002 I re-ran only the Validated Improvement variant on test_data.json, now with lesson_001 only (`code/hw9/part4_retest_validated.py`). The other three variants do not depend on this decision, so their results did not change.

```
=== 4.4 Final results on test_data.json (12 held-out questions) ===
Variant               | Correct | Incorrect | Avg attempts | Avg latency (ms)
------------------------------------------------------------------------------
Baseline              |       0 |        12 |         1.00 |           1306.7
Evaluator Repair      |      11 |         1 |         2.00 |           7714.5
Lesson Memory         |      11 |         1 |         1.67 |           6236.1
Validated Improvement |      11 |         1 |         1.75 |           6637.1

Lesson Memory = every saved lesson; Validated Improvement = kept lessons only (lesson_001)
saved /Users/samina/Documents/data260-4098/reports/hw09/raw/part4_test_final.json
((.venv) ) samina@Kazis-MacBook-Pro-2 data260-4098 % echo "Kazi Samina Maraj Mumu"
Kazi Samina Maraj Mumu
((.venv) ) samina@Kazis-MacBook-Pro-2 data260-4098 %
```

Figure 12. Final test results for the four variants (Validated Improvement re-run with lesson_001 only).

Variant	Correct answers	Incorrect answers	Average attempts	Average latency (ms)
Baseline	0	12	1.00	1306.7
Evaluator Repair	11	1	2.00	7714.5
Lesson Memory	11	1	1.67	6236.1
Validated Improvement	11	1	1.75	6637.1

- **Baseline** got everything wrong. Without the reference facts, the model guessed the agency rules (for example 24 hours instead of 20 minutes) and also answered the two out-of-domain questions. Its latency (1306.7 ms) does not include evaluator time, because the Baseline has no evaluator in its loop; the other three variants include it.
- **Evaluator Repair** fixed 11 questions on the second try. The one it missed (te04) said the follow-up note is due 48 hours after the incident is closed, but the rule says before it is closed. The Evaluator caught this, and the system stopped after 2 attempts.
- **Lesson Memory** uses every saved lesson, including the rejected lesson_002. It got 11 correct, and 4 questions passed on the first try (te01, te03, te06 with lesson_001 and te07 with lesson_002), so it needed 1.67 attempts on average.
- **Validated Improvement** uses only lesson_001. It also got 11 correct, with 3 first-try passes (1.75 attempts, 6637.1 ms). It lost the first-try pass on te07 because lesson_002 was removed. This is the trade-off of the validation step: a little slower on label questions, but the system does not keep a lesson that caused a previously passing answer to fail the automated evaluation.

VII. Explanation

- **Agent communication:** the Answering Agent sends its answer to the Evaluator, and the Evaluator sends back PASS or REVISE with feedback. They do not share anything else.
- **Lesson reuse:** lessons are saved in lessons.json and loaded when their keywords match a new question.
- **Stopping condition:** at most 2 answers per question. The loop stops when the answer passes or after the second attempt.
- **Validation:** a lesson is only used by the Validated Improvement variant if it passes the checks on non-test questions (the validation set, plus the extra check for lesson_002). The test set was only used for the final measurement.
- **Limitation:** lessons are matched by keywords, which misses related questions that use different words. For example, tr05 (Severity 1) made the same kind of mistake but did not load lesson_001 because the words did not match. Also, my Evaluator's refusal rule only recognizes certain phrases, so a correct refusal like "cannot be answered" can be marked wrong (x4 in the extra check).

AI Use

1. What I used AI for and what I did myself. I used Claude to help write the HW9 code (Redis counter, semantic cache, sweep and stale demo scripts, the two agents and the lesson system), the datasets and test cases, a run guide, and to help write this report. I ran everything myself on my Mac with Redis, my MySQL database and qwen3:8b, took the screenshots, checked the results and pushed the code.

2. One wrong AI output and one thing I checked myself. The first stale demo assumed VTA Route 522 had route code RT-003, based on the seed script. In my database RT-003 is VTA Route 22, so the demo changed the wrong route and the invalidation did not trigger (Figure 13). I checked the real route codes myself with a SQL query on s4098_rel.

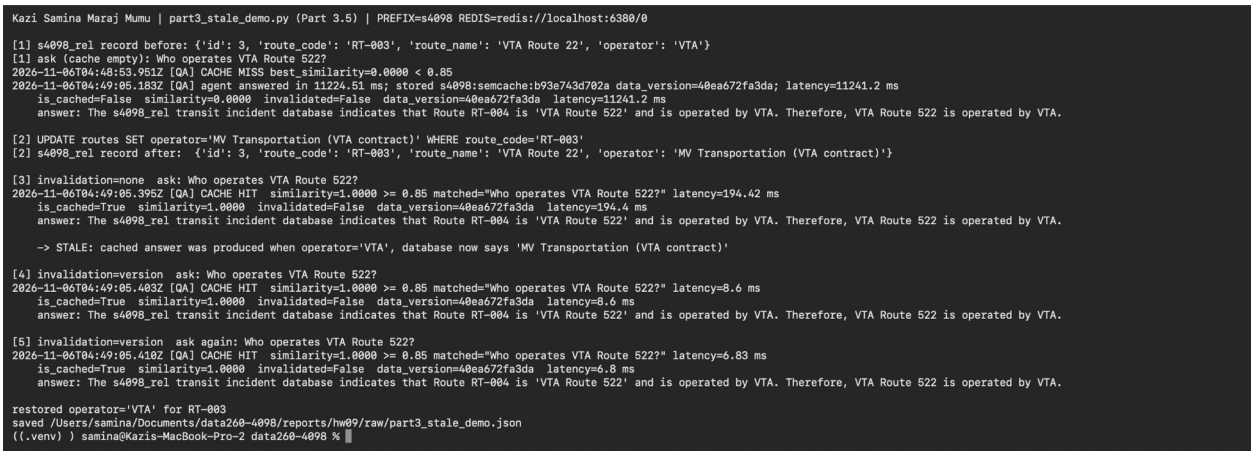


Figure 13. The first stale demo run, which updated the wrong route.

3. How I found it. In the output, step [2] showed 'VTA Route 22' even though the question was about Route 522, and the agent's own answer said Route 522 was RT-004.

4. What was fixed. The demo now finds the route by name instead of by code. After the fix, the data_version changed, the old entry was deleted and the new answer was returned (Figure 6). The reference fact for the Orange Line route code had the same problem, and I fixed it before running Part 4.

GitHub Repository

Repository: https://github.com/SaminaMaraj/data260-4098

Tag: hw9 (commit 6812411426deba9f0c65bab762dfcb13f1485119)